

EMCH 290

Mechanical Engineering Thermodynamics

Homework # 1

Points Possible: 30

Problem No. 1 (10 points)

An object whose weight is 100 lbf experiences a decrease in kinetic energy of 500 ft-lbf and an increase in potential energy of 1,500 ft-lbf. The initial velocity and elevation of the object, each relative to the surface of the earth, are 40 ft/s and 30 ft, respectively. If $g=32.2 \text{ ft/s}^2$, determine:

$$V_0 = 40 \text{ ft/s}$$

$$H_0 = 30 \text{ ft}$$

$$\text{Weight} = 100 \text{ lb}$$

$$\Delta KE = -500 \text{ ft} - \text{lbf}$$

$$\Delta PE = 1,500 \text{ ft} - \text{lbf}$$

$$KE = \frac{1}{2} * m * v^2$$

$$PE = m * g * h$$

(a) The final velocity, in ft/s.

$$-500 = \frac{1}{2} * \frac{100 \text{ lb}}{32.2 \text{ ft/s}^2} * (v^2 - 40^2) \rightarrow v = 35.749 \text{ ft/s}$$

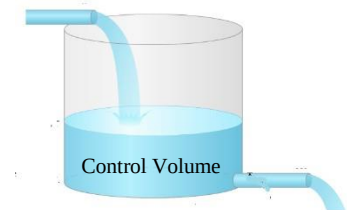
(b) The final elevation, in ft.

$$1500 = 100 \text{ lbf} * (H_1 - 30 \text{ ft}) \rightarrow H_1 = 45 \text{ ft}$$

Problem No. 2 (10 points)

Find the mass flow rate (in gram) of water for a control volume system, if the volume flow rate is 2 liter per second.

Volume flow rate = 2 (liter/s)
Mass flow rate =? (g)



Water density assumed to be $1g/cm^3$

$$1 \text{ liter} = 1000cm^3$$

$$2 \frac{\text{liters}}{\text{second}} * \frac{1000cm^3}{\text{liter}} = \frac{2000cm^3}{\text{second}} * \frac{1g}{cm^3} \rightarrow \frac{2000g}{\text{second}}$$

Problem No. 3 (10 points)

What is the change in the internal energy of the system if the surrounding releases 300 J of heat energy and if the system does 550 J of work on the surroundings?

Formula:

$$\Delta U = Q + w$$

Q = + endo

W = + on the system

Q = - exo

W = - by the system

550 J exothermic and 300 J “on the system”

$$\Delta U = 300 - 550 = -250 \text{ Joules, Loses energy}$$